

Choosing and Combining Biological Foundation Models

for Sequence Prediction

Literature Review

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Introduction and Review Scope

Our project asks how to choose biological inputs and pretrained models, and when combining their representations improves a defined prediction. DNA, RNA, and protein models offer different ways to represent related molecules, but their labels alone do not establish that fusion will help. The relevant questions are what information each input contains, what the encoder learned before the task, and whether the added representation improves a fair comparison. This review uses mRNA stability and transcript expression to connect those computational choices to the biological quantities they are intended to predict.

The review is a selective narrative synthesis. Its starting set is the eight papers on our research canvas, expanded with sources on codon regulation, sequence models, representation similarity, and evaluation. Relevant source passages and release documentation were checked where accessible; some checks relied on abstracts or indexed passages, as described in the disclosure. The review does not claim exhaustive coverage or that an untested comparison has never appeared elsewhere. Published papers support reported experiments, textbooks support established biology, and author documentation identifies software variants. Preprints and local project observations retain their separate status.

The organizing distinction is between biological information and access to that information through a learned representation. Additional genomic context can supply sequence absent from a transcript. A protein encoder receiving a translation of that transcript can instead contribute patterns learned from protein pretraining, even though translation has not created a new independent measurement. BioLangFusion, IsoFormer, LucaOne, and Central Dogma Transformer make different choices along these axes. Their results therefore inform different

parts of the project rather than forming a single performance ranking (Mollaysa et al. 2–3; Garau-Luis et al. 4–6; He et al. 943–45; Ota 7–9).

Biological Inputs and Prediction Targets

From Sequence Instructions to Cellular Function

The central dogma supplies an initial map of the sequence relationships. DNA and RNA consist of chains of nucleotides; the order of those building blocks carries sequence information. Transcription makes an RNA strand using DNA as the template: the DNA stays DNA. For a protein-coding message, a ribosome reads the mRNA and joins amino acids in the specified order, with transfer RNAs bringing those amino acids. This is translation. The message supplies instructions; the amino acids supply the material (Alberts et al., *Essential*, ch. 7). Amino acids are building blocks for proteins, the large molecules that do much of a cell's work (Nelson et al., ch. 3 introduction and sec. 3.1). A model accepting these sequences receives representations of related biological objects; it does not receive the physical machinery carrying out those processes.

Within a coding region, the standard genetic code maps groups of three nucleotides, or codons, to amino acids or stop signals. In our explanatory example, AUG–GCU–GAA specifies methionine–alanine–glutamate in that reading frame. GCU specifies alanine; it does not specify the entire preceding chain. Multiple codons can encode the same amino acid, so translating a coding sequence removes synonymous-codon identity (Nelson et al., sec. 27.1). This explains a concrete difference between modalities: two coding mRNAs can produce identical protein inputs while retaining different nucleotide inputs. Whether those differences help predict stability or expression must then be tested.

Inside a eukaryotic nucleus, DNA is organized with proteins into chromatin. DNA wraps around histone cores, like a rope around spools. Packing solves the storage problem, but the cell also needs access to the DNA. The DNA sequence carries genetic information; chromatin organization helps control access to it. Accessible DNA is not automatically being expressed: transcription also depends on regulatory proteins and the transcription machinery (Alberts et al., *Molecular* 189–90, 198, 204, 334).

The genomic region supplied to a model therefore matters as much as its DNA label. Regulatory DNA can occur near a promoter, far upstream, within introns, or downstream, and looping can bring bound regulators into contact with transcription machinery (Alberts et al., *Molecular* 334, 415). Converting a coding RNA sequence into DNA letters does not recover its surrounding genomic context. Conversely, supplying a genomic sequence window does not directly measure chromatin accessibility or regulator occupancy. These distinctions help explain why BioLangFusion’s sequence-derived views and IsoFormer’s genomic context are different modeling choices.

A mature transcript also differs from the genomic region that produces it. Splicing removes introns and joins exon sequences; alternative splice choices can produce related transcript isoforms. Exons are not synonymous with protein-coding sequence: untranslated regions remain in mature mRNAs and can regulate translation, localization, and persistence (Alberts et al., *Molecular* 339–40, 446–47, 455–56, 460). A full-transcript input can therefore retain information absent from a coding-sequence-only input or its translated protein. Cropping a transcript changes which regulatory information remains available. The significance of that choice depends on the target and experimental context.

RNA abundance, transcription rate, and RNA stability are related but distinct quantities. An mRNA's abundance depends on both its production and degradation; at steady state those rates balance (Nelson et al., sec. 26.2). An abundance measurement alone does not identify which rate caused it. Consequently, a transcript-expression benchmark and a stability benchmark answer different biological questions, even if some sequence features contribute to both. This distinction does not require assuming that every tissue sample obeys an identical steady-state kinetic model.

Amino acids have different side chains, and their order helps determine how a protein folds and functions. Many proteins act as enzymes: biological catalysts that speed reactions and can be used again afterward. Folding, chemical modifications, and cellular location also matter. For proteins entering the endoplasmic reticulum, helper proteins assist folding and many chains acquire sugars. Proteins transported onward to the Golgi undergo further processing and sorting for their destinations. A sequence embedding can help predict properties associated with these processes, but does not directly measure the protein's folding state, modifications, location, or activity in a particular cell (Nelson et al., secs. 3.1, 3.4, 4.4, 6.1–6.2; Alberts et al., *Molecular* 712, 714–15, 770–71).

Codon Effects and the Meaning of a Biological Measurement

Synonymous codons provide a concrete connection between sequence differences and biological outcomes. Presnyak et al. changed synonymous codons in yeast reporters and measured changes in mRNA decay and translation. Their experiments support codon optimality as a determinant of stability in that system. Optimality concerns how codons are decoded in a cellular context; it is not simply a universal ranking of frequently used codons. This evidence

explains why retaining nucleotide distinctions may matter to a stability predictor. It does not, by itself, demonstrate that adding a protein representation improves that predictor (figs. 4–5).

Saluki supplies a task-specific computational comparison. Agarwal and Kelley compiled 39 human and 27 mouse mRNA-decay datasets, examined technical biases, and trained a convolutional and recurrent predictor using transcript sequence, coding-frame annotations, and splice-site positions. The reported human Pearson correlation was 0.77 on consensus half-life targets under held-out evaluation. This demonstrates predictive signal under that protocol; it is not an upper bound on stability prediction. It also cannot be ranked directly against a model evaluated on different species, transcript regions, labels, or metrics (fig. 5; “mRNA Half-Life Data Collection and Pre-Processing”).

A different experiment measured protein production from 1,459 synonymous variants of a red fluorescent protein gene in *Escherichia coli*. Nieuwkoop et al. found strong predictive information near the translation start, with a window spanning codons 2–8 and a proposed role for local mRNA structure. The structure interpretation used computational predictions, and the result concerns this reporter, host, and design. Fluorescence-based protein production is not the same measurement as mRNA half-life (figs. 4–7). For our purposes, identical amino-acid inputs give a deterministic protein encoder identical representations, so that branch cannot distinguish these variants by sequence alone. Nucleotide representations can retain the differences. This is our interpretation of why the dataset provides a useful control; the project README already describes its use in Stage 1 (“Multimodal-bio-fusion,” sec. “Stage 1 Analysis”).

Sequence design offers a related application. Diez et al.’s iCodon builds on existing stability profiles and a prior predictor to search synonymous coding variants, with experimental testing in selected systems. Its stability score is not a direct duration in hours, and fluorescence,

decay, and expression measurements must remain distinct. The work provides an example of prediction informing experimental design; it does not establish that every optimized sequence will perform as predicted or that a foundation-model fusion method is required (figs. 1–4; “iCodon as an mRNA Stability Predictor Tool”).

Tissue-expression data introduce another biological context. GTEx maps associations between genetic variation and RNA expression or splicing across human tissues; these associations depend on tissue and cell composition. The full atlas should not be equated with IsoFormer’s selected 30-tissue benchmark. Nor does RNA expression directly measure protein activity. These distinctions determine what a model trained on GTEx-derived targets can plausibly contribute to a biological hypothesis (GTEx Consortium, “Tissue Specificity of Genetic Regulatory Effects”; Garau-Luis et al. 6–7, 16).

Choosing Pretrained Representations

Encoders and Their Training Objectives

An encoder converts a sequence into numerical representations that a predictor can use. Tokenization determines how the sequence letters are grouped into input units called tokens. A token can represent a nucleotide, several nucleotides, or an amino acid; it is a computational unit, not another molecular building block. An embedding is a learned numerical vector associated with an input unit or sequence. Attention allows a model to weight information from other represented positions; cross-attention extends that operation between streams of representations. Pooling summarizes multiple positions, and a prediction head maps the resulting features to a target. Pretraining supplies parameters learned before the downstream task. Freezing an encoder keeps those parameters fixed; fine-tuning updates them for the task. The encoder’s pretraining

task and the downstream prediction task should be identified separately (Mollaysa et al. 2–4; Garau-Luis et al. 4–6).

Many of the selected models use Transformer encoders and masked-token pretraining, which learns to recover hidden sequence units from context. The Transformer paper explains the attention architecture; the individual biological model papers establish their training objectives. HyenaDNA instead uses autoregressive next-nucleotide prediction with long convolutions, and Enformer learns supervised mappings from DNA to expression and chromatin-associated measurements. These objectives create different reasons for a representation to be useful. A model’s ability to accept a sequence does not establish that its pretraining or context window matches the intended task (Vaswani et al., sec. 3; Dalla-Torre et al. 288–89; Nguyen et al., “HyenaDNA,” secs. 3–4; Avsec et al., “Model Architecture”).

Table 1 identifies the twelve frozen encoders recorded for Stage 1. It separates the selected model variants from the kinds of sequences and tokens used in pretraining. These are not twelve independent model families: the selection includes two Nucleotide Transformer variants and three ESM-2 sizes. Literature-level comparisons explain why the models are candidates; the project’s recorded layer, pooling, sequence length, and checkpoint settings determine the actual extracted features (“Multimodal-bio-fusion,” sec. “Stage 1 Analysis”).

Table 1

Selected Encoders and Their Pretraining Inputs

Selected encoder	Pretraining sequence	Token unit	Role and qualification
NT 500M human-ref	Human reference genomic DNA	6-mers	Human-genome representation; CDS input omits surrounding regulation.
NT v2 100M multi-species	Genomic DNA from 850 species	6-mers	Broader genomic pretraining; differs from NT 500M in size and architecture too.
DNABERT-2 117M	Multispecies genomic DNA	Byte-pair segments	Variable-length DNA units; other model differences also matter in project comparisons.
HyenaDNA large	Human reference genomic DNA	Single nucleotides	Long-convolution model; long context does not guarantee distant regulatory insight.
RNA-FM	Non-coding RNA from RNACentral	Single nucleotides	RNA sequence representation; coding-region transfer requires evaluation.
RiNALMo mega (150M)	Non-coding RNA from several databases	Single nucleotides	Selected smaller variant; headline paper results emphasize the 650M model.

Selected encoder	Pretraining sequence	Token unit	Role and qualification
mRNA-FM	mRNA coding sequences (CDS)	In-frame codons	Coding-sequence model; release documentation supplies model-specific details.
CaLM (86M)	Protein-coding DNA from ENA	In-frame codons	Retains synonymous codon information; DNA-pretrained despite the project's RNA grouping.
ESM-2 8M, 35M, and 150M	Protein sequences from UniRef	Amino-acid residues	Three sizes of one family; identical protein sequences omit synonymous differences.
ProtBERT (UniRef100)	Protein sequences from UniRef100	Amino-acid residues	Different protein-model family; selected release is not ProtBERT-BFD.

Note. CDS means coding sequence. NT uses nonoverlapping six-mers with single-base fallback.

These are pretraining inputs; the project groups CaLM with RNA encoders although its original pretraining used protein-coding DNA. Model-family findings do not establish performance of every selected variant.

Sources: Dalla-Torre et al.; Zhou et al.; Nguyen et al., “HyenaDNA”; Chen et al.; Penić et al.; ml4bio; Outeiral and Deane; Lin et al.; Elnaggar et al.; Rostlab; “Multimodal-bio-fusion.”

For DNA representations, Nucleotide Transformer primarily groups six nucleotides into a token, while DNABERT-2 uses byte-pair segments and HyenaDNA uses individual nucleotides. DNABERT-2’s reported 36-dataset collection spans GUE and GUE+, which the methods distinguish. Its appendix also compares byte-pair and six-mer tokenization under matched training conditions, providing direct evidence about that choice within the tested configuration. HyenaDNA demonstrates very long sequence contexts, but its family’s maximum supported length does not establish the length used in a particular experiment. A coding sequence supplied in DNA letters also lacks the surrounding genomic context that could otherwise inform regulation (Dalla-Torre et al. 288–89; Zhou et al., secs. 3.1, 4.1–4.2 and app. A.6; Nguyen et al., “HyenaDNA,” secs. 3–4).

RNA and codon models differ in more than their molecular label. RNA-FM and RiNALMo were pretrained on noncoding RNA, while mRNA-FM and CaLM use coding sequences and codon tokens. RiNALMo’s reported generalization to new RNA families concerns families withheld during downstream fine-tuning; it should not be read as proof that no related family appeared anywhere in pretraining. Its 150M checkpoint is distinct from the paper’s principal 650M model. For mRNA-FM, the author release documentation supplies the model-specific description of coding-sequence and codon inputs (Chen et al., “RNA Foundation Model Training Details”; Penić et al., fig. 1 and Methods; ml4bio; Outeiral and Deane, Methods).

CodonBERT is relevant as a codon-level model and as the source of benchmarks used by BioLangFusion. Its pretraining combines masked codon prediction with taxonomy-related learning, and its evaluated tasks include expression, stability, and degradation. CaLM also demonstrates useful codon representations on sequence-property tasks. These studies motivate retaining synonymous-codon distinctions, but cross-model comparisons change training data,

architecture, scale, and sometimes retained input information together. They do not isolate tokenization as the sole cause of a performance difference. An expression experiment involving candidate vaccine mRNAs likewise does not establish clinical vaccine efficacy (S. Li et al.; Outeiral and Deane, figs. 2–3 and Methods).

Lin et al. train ESM-2 to recover masked amino acids from protein sequence context. The study examines models from 8 million to 15 billion parameters and finds that structural information becomes more accessible as scale increases, with performance also depending on the evolutionary relatives represented in training. Its contact-prediction experiments test a specific connection between learned sequence patterns and protein structure (fig. 1).

ESM-2 must also be distinguished from ESMFold. ESMFold adds a folding network that turns language-model representations into predicted atomic coordinates. Reusing ESM-2 features in an expression predictor does not mean that the predictor runs this folding network or measures protein activity. The distinction connects the textbook’s sequence–structure relationship to what the actual computational pipeline contains (Lin et al., fig. 2; Nelson et al., secs. 3.4, 4.4).

ProtBERT contributes another protein-pretraining approach. The ProtTrans study compares several architectures and corpora, whereas the selected ProtBERT release identifies UniRef100 as its training source. Family-wide corpus sizes and best downstream results therefore should not be attributed automatically to this checkpoint. As with ESM-2, a downstream predictor must extract the information relevant to the target; the pretrained representation is not itself a measurement of molecular function (Elnaggar et al.; Rostlab).

Layer Choice and the Limits of Model Comparisons

The Nucleotide Transformer layer comparison is directly relevant to our proposed project: among the probed configurations, the final layer never produced the highest

performance. This is evidence from their genomic benchmarks, not a guarantee about mRNA stability or all encoders. The authors also report improved results after task adaptation and distinguish the later NT-v2 models from the original series. Our inference is that layer selection deserves a controlled experiment rather than an assumption that the deepest representation is best (Dalla-Torre et al. 288, 294–95).

Independent benchmarking provides an additional check on model-release claims. BEND evaluates frozen DNA representations with downstream predictors on biologically defined tasks and finds that useful transfer varies substantially, especially for long-range problems. Its findings apply to the tested extraction and prediction protocol, rather than imposing a ceiling on every fine-tuned version of a model. Together with the layer comparison, this supports choosing encoders through matched task evaluation rather than parameter count or modality name alone (Marin et al., secs. 3–6).

Alignment and Fusion as Different Questions

What It Means for Representations to Agree

Biological correspondence links a transcript with its genomic origin and, when applicable, its protein product. Positional correspondence links sequence locations, such as a codon and its encoded amino acid. Geometric similarity asks whether two representations organize matched examples similarly. Alignment can also name a training objective intended to bring paired examples together. These relationships should be specified separately: knowing that sequences correspond does not establish that their embedding coordinates share a meaning, and useful fusion need not maximize a similarity score (Mollaysa et al. 2–3; Garau-Luis et al. 4–5; Kornblith et al., sec. 3; Radford et al., sec. 2.2).

Centered kernel alignment, or CKA, compares the relationships among matched examples in two representation sets. A neighborhood measure instead asks whether the same examples appear close together in both spaces. Huh et al. use mutual nearest-neighbor overlap to examine convergence across model representations, while acknowledging that modalities can contain different information. These measures describe geometry under their chosen assumptions. Neither high CKA nor neighborhood agreement alone establishes that two encoders contain equivalent information or that combining them will improve a biological prediction (Kornblith et al., secs. 2–3; Huh et al., app. A and sec. 6).

A probe asks a different question: how well can a specified predictor recover a target from a frozen representation? Linear probes test what is accessible through a linear mapping; they do not measure all information a more flexible model might recover. Hewitt and Liang show why probe capacity and suitable control tasks matter when interpreting such results. Applying that principle to biological sequences, length and composition are useful inexpensive controls. Their predictive value is biological signal too; the question is what the learned representation adds beyond those selected features (Alain and Bengio, sec. 3; Hewitt and Liang, secs. 2–3).

Tjandrasuwita et al. investigate when geometric alignment accompanies predictive performance using controlled synthetic data and nonbiological multimodal tasks. Greater task-relevant uniqueness can reduce observed alignment, and better prediction need not produce more similar representations. In a movie-genre experiment, varying a contrastive alignment loss produces different effects across categories; the evaluated prediction uses the language branch. This is evidence that an alignment objective can help or hurt under particular conditions, rather than a direct biological test of fused-versus-single-encoder performance (3–9).

The implication for our project is conditional. When protein sequence is derived from a coding RNA, it cannot add synonymous-codon identity absent from that translation, although protein pretraining may make useful sequence constraints easier for a predictor to extract. When a DNA branch instead receives a genomic window, it can contain sequence outside the transcript. Complementarity therefore depends on actual inputs, target, and encoders. Residual similarity after controlling selected compositional features should be reported alongside held-out prediction and fusion ablations; it is not proof that a molecular mechanism has been isolated.

Ways of Combining Modalities

Multimodal surveys distinguish combining input features, allowing intermediate representations to interact, and combining final predictions. These are useful descriptions of where fusion occurs, but the actual operations and training settings should be stated. Concatenation places feature vectors together; attention can learn interactions among represented positions; and contrastive training can encourage corresponding examples to occupy compatible spaces. CLIP provides an example of that last objective for images and text. Its success motivates a possible method, not a guarantee of benefit for frozen biological encoders (Baltrušaitis et al., sec. 6; Radford et al., sec. 2.2).

BioLangFusion begins with mRNA sequences and property labels, deriving corresponding DNA and protein sequences. Its encoders are Nucleotide Transformer v2 for DNA, RNA-FM for RNA, and the 8-million-parameter ESM-2 model for protein. Because their token units differ, the authors bring their representations onto a common grid referenced to protein positions: RNA representations are averaged in groups of three, and a learned transposed convolution expands the DNA representation. This implements the intended coding correspondence. The precise treatment of untranslated regions, reading frames, and special

tokens requires further clarification before assuming that the same mapping is valid for every retained example (Mollaysa et al. 2, 7–8).

The paper compares three ways of combining those representations. Concatenation places features from the modalities together at corresponding positions. Attention pooling inspired by multiple-instance learning assigns a weight to each modality for a sequence and uses those weights to combine the position-preserving representations. Its entropy penalty encourages more concentrated weights. Cross-attention allows each modality to use the combined context from all three streams, including other positions. A convolutional prediction head then produces a task-specific output. These choices encode different assumptions about how information should be combined, so greater architectural complexity does not itself imply a better prediction (Mollaysa et al. 2–4, 9–11).

BioLangFusion is presented as reusing pretrained base models without additional base-model pretraining or modification. The downstream projections, fusion components, and predictor are trained. Reusing pretrained features therefore does not mean that nothing is trained; the paper does not explicitly document all encoder gradient settings. A further conceptual limit is that deriving three sequence views from an mRNA does not create three independent molecular measurements. Complementarity can instead arise from the different information retained by the sequences and the different patterns their encoders learned during pretraining (Mollaysa et al. 1–3).

IsoFormer supplies a different combination of inputs: genomic DNA around a transcript's transcription start site, the transcript sequence, and its associated protein when one exists. The principal comparisons use Nucleotide Transformer v2 or Enformer for DNA, Nucleotide Transformer v2 for RNA after replacing U with T, and ESM2-150M for protein. The RNA input

includes untranslated regions; long transcripts are cropped from the left to retain the 3' end. The Enformer DNA window is 196,608 nucleotides, compared with 12,288 for the Nucleotide Transformer configuration. Thus, the DNA branch can carry genomic context absent from a mature RNA sequence, and the encoder comparison changes the available context as well as the model (Garau-Luis et al. 4–6, 17).

IsoFormer uses successive cross-attention operations and residual connections to let each stream incorporate information from the others. Its aggregation and prediction stages yield one expression prediction per tissue. The pretrained encoders are jointly fine-tuned with the fusion module and head. For transcripts without an associated protein, missing-modality cross-attention contributions are set to zero. This accommodates noncoding transcripts computationally, although the paper does not separately establish their predictive performance. Unlike BioLangFusion's codon grid, the architecture does not require every genomic position to have a directly corresponding amino acid (Garau-Luis et al. 4–6).

Coupled Mamba provides a state-space alternative. Each modality maintains a learned state, and updates incorporate previous states from the other streams. The published study includes breast-cancer subtype classification from measured mRNA expression, DNA methylation, and microRNA expression, alongside multimedia tasks. A methylation profile records chemical marks on DNA; a microRNA expression profile records the abundance of small regulatory RNAs, rather than another coding-sequence view (Alberts et al., *Molecular* 435–37, 463–64). Coupled Mamba reports 88.1% accuracy on the BRCA benchmark, compared with 87.5% for MM-Dynamics. Those inputs are molecular measurements, not DNA–RNA–protein sequence embeddings. The paper also tests temporally unaligned multimedia, so its

computational steps should not be mistaken for demonstrated biological position matching (W. Li et al. 3–6, 8–10).

Adapting that architecture to the project requires deciding what each state update indexes, how features are ordered, and how branches with different lengths or genomic context interact. A codon grid is one possible construction for corresponding coding sequences, but it does not align every genomic position with a protein residue. Efficiency comparisons also depend on model size, sequence length, hardware, and implementation. Coupled Mamba supplies a candidate for matched testing; its benchmark results do not settle those design choices or establish clinical usefulness (W. Li et al. 5–10).

Shared Pretraining and Directional Architectures

LucaOne asks whether a shared encoder can learn useful nucleic-acid and protein representations during pretraining. He et al. mix the sequence types without explicit DNA–protein pairing labels. Their 1.8-billion-parameter transformer uses nucleotide and amino-acid tokens, identifies molecular type, and combines masked-token prediction with annotation objectives, including protein structural information. Its learning signal therefore extends beyond raw sequence reconstruction. This differs from BioLangFusion’s combination of independently pretrained encoders and IsoFormer’s downstream joint fine-tuning (943–44, 948–49).

Its central-dogma benchmark classifies whether DNA and protein sequences form a correct pair. The frozen encoder supplies pooled representations that are concatenated for a trained classifier. The original benchmark reports accuracy of 0.8453, compared with 0.8048 for separately pretrained gene/protein versions and 0.7309 for a DNABERT2 plus ESM2-3B combination. The dataset uses constructed negative pairs; the task is not protein generation or a simulation of gene expression. Harder homologous-pair experiments in the supplement are

sensitive to the training allocation, further qualifying generalization from the original benchmark (He et al. 944–45, 950; extended data tables 2, 4–5; supplementary note 5).

Evo offers another route to tasks spanning molecular levels, but its pretraining input is genomic DNA. It uses next-token sequence modeling with a StripedHyena architecture and supports predictions and selected experimentally tested designs involving nucleic acids and proteins. That construction differs from LucaOne’s mixed molecular inputs. Neither a shared-model label nor a central-dogma label determines the information supplied to the model; the tokenizer, training examples, and evaluation must be examined directly (Nguyen et al., “Sequence Modeling,” Discussion and Materials and Methods).

Central Dogma Transformer instead imposes an ordered exchange between representations: RNA queries DNA, then protein queries the DNA-enriched RNA stream. Ota’s version 2 manuscript describes the implementation called CDT v1. Its biological inputs need close inspection. Frozen Enformer processes enhancer-centered DNA; the RNA stream uses static scGPT gene-token lookup embeddings without expression values or contextualized scGPT outputs; fixed protein representations come from ProteomeLM. RNA and protein rows are matched by gene identity. The same gene/protein matrices are shared across examples, with enhancer DNA providing the varying molecular input. Calling a stream RNA therefore does not imply that it contains an RNA sequence or a sample-specific transcriptomic measurement (7–9, 29–30).

The demonstrated task predicts the effects of CRISPRi enhancer perturbations on gene expression in K562 cells. CRISPRi is used here to repress an enhancer and measure the associated expression response; the study reuses an existing screen. The manuscript reports a Pearson correlation of 0.503 on the validation set used for early stopping. That result is not an

independent test score, and the reported evidence does not establish an advantage over unimodal or alternative-direction baselines. Ordered attention is consequently a biological design assumption whose value still needs a controlled comparison (Ota 12–13, 28, 30–33).

A highlighted FNDC5 example connects model attribution to an annotated CTCF site and existing chromatin-contact evidence. Such a correspondence can motivate a specific hypothesis, but the author acknowledges that establishing CTCF-mediated causation would require an appropriate perturbation. The pipeline also sends several streams directly to the final representation, so output influence need not traverse the entire proposed sequence of interactions. CDT is useful as a hypothesis-generating proposal; its name and directionality should not be treated as proof that the model reproduces transcription, translation, or cellular mechanisms (Ota 9, 15–18, 22–24).

Evidence and Limits of Interpretation

Reported Gains and Biological Endpoints

BioLangFusion evaluates five tasks described as vaccine-associated mRNA degradation, fungal expression, *Escherichia coli* protein-expression categories, transcript stability, and expression from antibody-encoding mRNAs. Regression performance is reported using Spearman correlation, which assesses agreement in ranking; *E. coli* classification uses accuracy. The targets must retain their assay-specific meaning. An antibody-expression prediction is not a prediction of binding affinity, and an mRNA-degradation score is not a measurement of vaccine efficacy (Mollaysa et al. 4, 7–8).

At least one fusion variant exceeds the strongest listed unimodal score on each task in BioLangFusion’s Table 1, but the winning method varies. For transcript stability, attention

pooling reports a Spearman correlation of 0.563, compared with RNA-FM's 0.553.

Concatenation and cross-attention report 0.539 and 0.550, respectively, below that unimodal baseline. Cross-attention leads the main comparison for *E. coli* classification and antibody expression. The evidence therefore favors task-dependent testing rather than a universal preference for fusion or cross-attention. Because the table supplies no repeated-seed variability or uncertainty intervals, the 0.010 stability advantage remains a reported numerical difference whose reliability needs further evaluation (Mollaysa et al. 4).

IsoFormer predicts transcript abundance across 30 tissues using GTEx v8 expression measurements averaged across individuals. The labels are log-transformed and normalized by tissue before training. They represent tissue-average RNA abundance rather than protein concentration, an individual patient's expression, or single-cell activity. Training and testing are separated by gene, which helps prevent different isoforms of one gene from appearing in both partitions. The reported means and standard deviations summarize five seeds on a fixed test split (Garau-Luis et al. 6–7, 16).

Its modality ablation supplies direct evidence of benefit within one configuration. With Nucleotide Transformer encoders, RNA alone achieves an R^2 of 0.36 ± 0.03 ; DNA plus RNA reaches 0.39 ± 0.01 ; and all three modalities reach 0.43 ± 0.01 . R^2 evaluates predictive fit relative to a mean-based reference, rather than the ranking measured by Spearman correlation. With Enformer in the DNA branch, the multimodal model reaches 0.53 ± 0.01 . This latter comparison changes encoder architecture, pretraining, and sequence context together, so it cannot isolate which change caused the improvement (Garau-Luis et al. 7–8).

The size of an added modality's contribution also depends on the configuration. The appendix reports R^2 values of 0.52 for Enformer plus RNA and 0.53 after adding protein.

Furthermore, cross-attention and the alternative C-Abstractor aggregation module share the headline R^2 of 0.53 ± 0.01 . These results support testing the contribution of each component, rather than attributing the full multimodal advantage to protein input or one fusion mechanism. BioLangFusion's stability correlations and IsoFormer's expression R^2 values cannot rank the two systems against one another because their tasks, datasets, targets, and evaluation procedures differ (Garau-Luis et al. 9, 18; Mollaysa et al. 4).

The broader evidence also resists a universal ranking. LucaOne evaluates several molecular targets, including an influenza benchmark based on antigenic relationships between haemagglutinin RNA sequences, but does not lead every reported task. Such an endpoint is different from prospective vaccine effectiveness. Similarly, Coupled Mamba's measured-omics subtype classification and ESMFold's structural predictions address different quantities from mRNA decay or transcript abundance. A biological application should therefore begin with the measurement the model was evaluated against, rather than borrowing significance from another task (He et al. 947, 950–51 and extended data table 5; W. Li et al. 6–8; Lin et al., fig. 2).

Controls That Make a Comparison Interpretable

Leakage and inconsistent evaluation can make apparent improvements difficult to interpret. Kapoor and Narayanan document such problems across machine-learning-based science; their work motivates explicit checks, but does not establish that any particular biological paper here contains leakage. For sequence data, duplicate records, related sequences, preprocessing learned from held-out data, and overlapping biological groups require attention. The appropriate split depends on whether the intended prediction concerns new transcripts, genes, families, tissues, or experimental conditions (Kapoor and Narayanan).

The papers differ in how completely their descriptions support reproduction.

BioLangFusion gives an intended common prediction-head design, but the appendix lists different convolution-channel counts for its attention-pooling and cross-attention examples. Its DNA alignment convolution also differs between the main equation and appendix listings. Those discrepancies should be resolved against a versioned implementation before treating the fusion comparison as perfectly controlled. They are reporting limitations, not evidence that the reported scores are false (Mollaysa et al. 2, 4, 9–11).

IsoFormer reports gene-separated testing but leaves several preprocessing and evaluation details unspecified, including the exact train/test counts and the population used to fit the tissue-normalization statistics. It also does not establish a chromosome-held-out or sequence-homology-controlled evaluation. These are unresolved checks on the scope of generalization, not proof of leakage. Five-seed variation on one test split addresses a different uncertainty from performance on new biological cohorts or experimental conditions (Garau-Luis et al. 6–7, 16).

A matched comparison should keep examples, targets, splits, metrics, and tuning opportunities comparable while changing the component of interest. Sequence length, nucleotide or codon composition, the strongest relevant single encoder, and simple feature concatenation provide distinct baselines. Repeated training seeds assess optimization variability; alternative partitions assess variability associated with the sampled split. Neither alone establishes generalization to a new cohort or assay. These are recommendations drawn from the review's comparison, not evidence that every listed control has already been run in each paper.

BioLangFusion and IsoFormer also require care when connecting attention to biological explanation. A modality weight is a coefficient applied to that modality's representation in a weighted sum; its size does not by itself measure unique information or predictive contribution.

A change in attention toward an RNA region can suggest a feature worth examining. Neither observation alone establishes that the region or modality causes the predicted biological outcome. Mechanistic interpretation would require additional evidence, such as controlled perturbations and measurements that distinguish competing explanations (Mollaysa et al. 3, 9–10; Garau-Luis et al. 8–9, 16–17).

Implications for the Research Project

The literature supports a sequence of decisions. First specify the biological target and how it was measured. Then identify whether each proposed branch supplies additional sequence context, a different molecular measurement, or another learned representation of related information. Select encoders and layers under a common evaluation before interpreting a more elaborate fusion architecture. The inspected papers leave open a matched comparison of the project’s chosen encoders, alignment measures, and fusion methods; this is a bounded research question rather than a claim that no biological study has ever investigated their relationship.

The current repository records Stage 1 analyses of twelve frozen encoders on 981 retained stability sequences and a GTEx pilot. On the GTEx train pilot, four transcript-length features achieve a cross-validation R^2 of 0.242, compared with 0.239 for all twelve embeddings combined. These are similar observed scores, not a formal equivalence result or the published-test evaluation. Composition controls also substantially reduce similarity involving DNA encoders. These observations are reported project findings, not results independently rerun for this review, and they concern pooled representations with linear probes rather than reproductions of the trained BioLangFusion or IsoFormer architectures (“Multimodal-bio-fusion,” sec. “Stage 1 Results”).

The repository's Stage 2 description treats alignment as agreement between representations after controlling for selected sequence composition and reports it separately from predictive complementarity. It records a decision to begin without an explicit alignment loss. These are operational choices for the project, not universal rules. Token-level fusion is planned for Stage 3. The recorded architecture priorities put Coupled Mamba first, with IsoFormer-style cross-attention as another comparison and concatenation plus an MLP as a reference baseline. These priorities explain what the literature review should help evaluate; they do not establish which method will work best ("Multimodal-bio-fusion," secs. "Architecture Priorities," "Stage 2: Defining Alignment," and "Next Steps: Stage 3 Fusion Experiments").

A useful next experiment can therefore ask whether a measured relationship between two frozen encoders predicts an added benefit from combining them under matched conditions. Comparing several encoders within a modality can test whether a conclusion survives model choice, while causal claims about tokenization or pretraining require more controlled interventions. Another proposed question is whether an earlier encoder layer preserves predictive performance within a predefined tolerance. A small observed difference is not sufficient to establish that tolerance or equivalence (Dalla-Torre et al. 288, 294–95; Tjandrasuwita et al. 8–9). This earlier-layer extension and the broader research-question expansion remain proposals for advisor discussion.

Biological follow-up should be equally specific. A stability predictor could help prioritize synonymous coding variants for decay measurements; an isoform-expression predictor could nominate transcript differences or sequence regions for testing. Protein production, mRNA half-life, antigenic similarity, and protein conformation would each require their own measurements. The experimental codon studies demonstrate that computationally informed questions can be

tested in selected systems, while the foundation-model studies establish narrower predictive results. Combining those lessons gives the project a way to formulate hypotheses without presenting model attribution or a benchmark improvement as an established cellular mechanism (Presnyak et al., figs. 4–5; Diez et al., figs. 1–4; Nieuwkoop et al., figs. 4–7).

The central conclusion is that choosing modalities and choosing representations are connected decisions. Biological knowledge explains which distinctions can matter; empirical comparison establishes whether a particular model uses them productively. The contribution to pursue is a defensible method for choosing and testing those combinations, with the target, evidence, uncertainty, and possible experimental use kept visible throughout.

Preparation and AI Assistance

Chase Di Maria Breisinger used Codex (OpenAI) to prepare and revise this literature review. He supplied original biology notes, earlier AI-refined versions, three biology textbooks, and project materials, and identified genome.gov before this review. His notes shaped explanations of sequence instructions, amino-acid building blocks, chromatin access, enzymes, and cellular processing. He directed the biological interpretation, source verification, integration of the authors' contributions, and academic formatting. Codex read the notes, relevant textbook sections, and project passages, retrieved additional sources, generated substantial prose and cross-paper synthesis, and revised the structure, citations, and presentation. Genome.gov remains part of the preparation history; the scientific statements retained here are supported by the cited books and papers.

Julian McKinley contributed research and writing prepared with Claude (Anthropic) to this literature review. The assistance information he provided reports AI support with finding additional sources, drafting, and formatting; checking publication details against Crossref, Europe PMC, publishers, and proceedings; checking added claims against available abstracts or documentation; and comparing the eight original paper summaries with annotations and project statements with the README. His Claude-assisted contribution informed the discussion of codon regulation, the encoder table, representation-similarity methods, and evaluation controls. The record does not establish which additional sources Julian personally selected or verified, or who first discovered every individual reference.

Codex integrated the authors' contributions and performed targeted source checks during revision of this review. These led to distinctions between model families and selected checkpoints, corrections to the treatment of Evo and mRNA-FM, narrower research-gap claims,

and qualifications on metrics and pilot results. Relevant full-text passages were inspected where available; some checks used author manuscripts, abstracts, indexed primary-text passages, or release documentation rather than a complete published-paper reading. Citations identify underlying evidence, while this disclosure records the assistants' influence. Most drafting and synthesis in this document was performed by AI. No experiments or notebooks were rerun for the review, and independent verification of the final prose, references, and ethical considerations by both authors is not yet documented.

The Codex conversation used to prepare this review is listed as a preparation source ("Codex-Assisted Literature-Review Preparation").

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